TOUSIF RAHMAN ANTO

Backend & Full-Stack Developer

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Scan for portfolio



Summary

I'm a Computer Science and Engineering graduate who builds backend and full-stack applications with Node.js, Django, React and Next.js. My projects include hospital billing software, a code conversion tool and a tournament platform. I work on database design, secure APIs, payments and deployment. I'm looking for an internship or junior engineering role where I can contribute to a development team.

Education

BRAC University Graduated September 2026
BSc in Computer Science & Engineering Dhaka, Bangladesh
CGPA: 3.00
Coursework: Data Structures & Algorithms, Object-Oriented Programming, Databases, Operating Systems, Software Engineering and Web Development.

Technical skills

Languages: JavaScript, TypeScript, Python
Backend: Node.js, Express, Django REST Framework, REST APIs, Socket.IO
Frontend & databases: React, Next.js, Tailwind CSS, MongoDB, PostgreSQL, Supabase
Tools & integration: Git, Docker, Postman, Vercel, Railway, OpenAI/Groq/Gemini APIs

Selected projects

Hospital Billing: Payments & settlement GitHub | Demo
React, Node.js, Express, MongoDB

- Designed a six-page data model and 16-screen wireframe set; built patient, encounter and invoice modules across three release milestones.
- Added staff permissions, charge capture and immutable invoice snapshots; used MongoDB Decimal128 for exact financial calculations.
- Integrated SSLCOMMERZ and Bangla QR for partial payments, refunds and reconciliation, with duplicate-safe settlement, PDF receipts, CSV exports and audit logs.

CodeSwitch: Code conversion across 5 languages GitHub | Demo
Python, Django REST Framework, React, TypeScript, OpenAI, Groq, Gemini

- Built code conversion with automatic switching between 3 AI providers during outages or rate limits and a rule-based fallback.
- Implemented a Django REST API with httpOnly JWT cookies, login-attempt lockouts, API-key rotation and Content Security Policy middleware.
- Added route-level code splitting and a lazy-loaded Monaco editor; deployed the backend to Railway and the frontend to Vercel.

AFK Productions Tournament Platform GitHub
Node.js, Express, MongoDB, React, Socket.IO

- Led development of a tournament platform used to coordinate a 25+ person production crew.
- Built JWT-authenticated REST APIs and permissions for administrators, players, team managers and sponsors; added Socket.IO notifications and live updates.
- Implemented brackets for single-elimination, double-elimination and round-robin tournaments, with validation and unit tests for bracket generation and authentication.

Projects continued

Game Physics Lab: Interactive 3D simulation GitHub | Demo

JavaScript, React, Three.js, Cannon-es, Vite

- Built a browser-based physics sandbox prototype with 11 simulation modes and six material presets, including collision, buoyancy, ramp and explosion experiments.
- Added GLB, GLTF and OBJ model imports with automatic collision bounds; implemented object selection, movement, rotation and scaling.
- Recorded peak speed, height, travel distance and estimated impact force with CSV/JSON exports; added automated tests for result calculations, scene records and simulation settings.

The Archive: Cloud document platform GitHub | Demo

Next.js, TypeScript, Supabase Auth & Storage

- Designed the Supabase database schema and sign-in flow, then built and deployed the Next.js application to Vercel.
- Implemented authenticated uploads with per-user storage isolation and in-browser PDF previews so users can read documents without downloading them.

Research

Project Lead: Local emotional-support conversational AI 2025 - 2026

Undergraduate thesis | Python, QLoRA, Mistral, Gradio, MongoDB

- Led a five-member team developing a non-clinical emotional-support chatbot. The contribution was an integrated local framework combining adapted language models, inspectable response checks, bounded correction and conversational memory.
- Implemented four interpretable indicators for empathy consistency, response drift, support quality and support consistency; connected response checks to guided regeneration and fallback handling before delivery.
- Prepared a dialogue-data pipeline and integrated Mistral, Qwen, DeepSeek and Llama for comparative work; used 4-bit QLoRA for local adaptation of 7B–8B models on a 12 GB consumer GPU.
- Integrated a Gradio interface, optional MongoDB memory, user ratings and identifier redaction. Accepted interactions informed later prompt guidance and summaries while model weights remained fixed during conversations.

Teaching experience

Dreamers Academy: Coding Instructor & Curriculum Developer 2024 - 2026

- Taught Python syntax, control flow, functions and structured problem-solving; provided individual learner feedback.
- Developed a Lua curriculum for Roblox Studio game development.

Leadership & service

Millennium Fellowship: Fellow, Class of 2025; Global Admissions Committee
Selected from a worldwide pool of 60,396 applicants. Led a student well-being project and reviewed fellowship applications using a scoring rubric. Received a letter of recommendation from the MCN Admissions Manager. Fellowship certificate | MCN recommendation letter

AFK Productions: Head of Human Resources 2024 - 2025
Coordinated recruitment, scheduling, communications and event logistics for a 25+ member esports crew.

Robotics Club of BRAC University: Human Resources Team 2024 - 2025
Supported membership, recruitment, workshops and campus robotics events.